

Garbage Classification

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1 Introduction

Waste Management Challenge

The United States is witnessing an exponential increase in waste production, with the average individual producing nearly 1,000 pounds of waste each year. This growth poses significant challenges to existing waste management infrastructure, which is burdened with the responsibility of sorting and processing diverse waste streams. The current system's limitations have led to inefficient waste processing, contributing to environmental pollution and resource depletion.

Deficiencies in Current Sorting Mechanisms

The conventional approach to waste segregation, heavily reliant on manual labor, is plagued by inaccuracies and inconsistencies. This outdated method results in a plethora of issues, including increased contamination rates, reduced recycling efficiency, and environmental hazards.

Project Objective

Our project addresses these challenges by introducing a machine learning model capable of automating the waste sorting process. By employing advanced computational techniques to classify waste accurately, we aim to enhance the efficiency and reliability of waste management systems.

Dataset Overview

Our ML model is trained on a comprehensive dataset provided by user cchangcs on Kaggle, containing 2467 images across six waste categories: Cardboard, Glass, Metal, Paper, Plastic, and Miscellaneous Trash. The images exhibit diverse characteristics, such as varied colors, patterns, and

lighting conditions, essential for training a robust classifier.

Significance of Automation

Implementing an automated sorting system is a pivotal advancement with cross-sectoral benefits:

- **Waste Management:** Streamlines sorting at waste facilities, improving recycling rates and reducing landfill reliance.
- **Environmental Monitoring:** Assists in assessing pollution levels, essential for environmental conservation efforts.
- **Urban Planning:** Facilitates optimized garbage collection routes, enhancing the efficiency of urban sanitation services.

Goals of the Project

Our project is structured around the following core objectives:

- **Data Preparation:** Ensuring the dataset's integrity and balance to support effective model training.
- **Model Development:** Leveraging convolutional neural networks (CNNs) and other deep learning algorithms to construct a state-of-the-art waste classification system.
- **Performance Evaluation:** Utilizing metrics such as accuracy, precision, recall, and F1-score to rigorously test the model. Cross-validation will be employed to ascertain the model's generalization capacity and to detect any overfitting or underfitting.
- **User Interface Creation:** Developing a web-based interface that allows for easy user interaction with the classification system.

- **Documentation:** Producing detailed documentation outlining the methodology, model architecture, and performance evaluations for use by practitioners and researchers.

Structure of the Report

This report will detail the identified problem, the innovative methodology proposed to address it, and the significant implications of the successful implementation of a garbage classification system. Subsequent sections will provide in-depth discussions on literature review, dataset analysis, proposed methodology, experimental results, and conclusions, supporting our project’s narrative and technical contributions.

2 Literature Review

Proper classification of disparate types of trash remains a problem that often goes overlooked, despite potentially massive implications for our waste management system. The EPA has estimated that roughly 32% of municipal solid waste ends up being recycled or composted, even though recyclables comprising around 75% of overall solid waste. Millions of tons of reusable materials per year end up being needlessly wasted due to a culture of carelessness and confusion surrounding refuse. Because of this, many cohorts of data scientists have produced various ML models with the goal of identifying potential recyclables that have been wrongfully condemned to landfills. Numerous such attempts have shown immense promise, as in the case of Aral et al. (2018), who demonstrated a 95% accuracy of identification through the use of deep convolutional networks such as Inception-V4. Their models were trained on a database of images containing six distinct types of refuse: paper, glass, metal, plastic, cardboard, as well as a catch-all class for miscellaneous non-recyclables. To demonstrate the efficacy of their models, tools such as Tensorflow and Keras were used to compare and contrast between their various implementations. They also found that the optimal range of epochs was between 100 and 200 depending on the model type. In another trial by Özkaya and Seyfi (2019), the power of tools such

as Support Vector Machines (SVMs) and Soft-Max regression was elucidated, eventually leading to a 97.86% accuracy via GoogleNet+SVM. Despite the success of such methods, both groups stressed the potential flaws inherent in their models, including the necessity of massive datasets for proper training, and their reliance on contrasting backgrounds potentially limiting their use in real-time systems.

3 Dataset Description and Exploratory Data Analysis

For this project we will be classifying images into 6 different categories: paper, plastic, trash, glass, metal, and cardboard. The dataset consists of 2527 samples. The distribution of classes is as shown below.

Each image has dimensions (584, 312, 3). Pa-

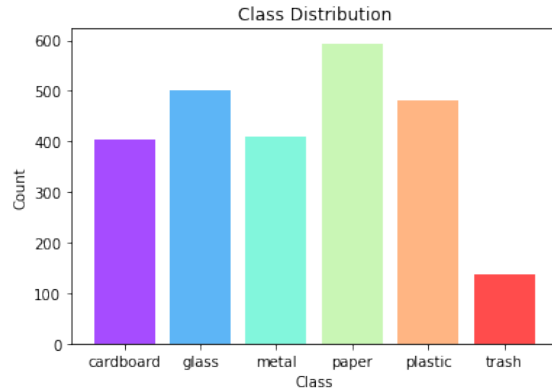


Figure 1: Distribution of samples in dataset



1.

Figure 2: Example photos of each class

per samples are most abundant with 594 samples, while trash samples are the least abundant with 137 total samples. The other images are fairly evenly distributed, however the slight variations in distribution may affect the results. For

the Support Vector Machine (SVM) and Random Forest Classifier, the images will undergo rescaling to size 128x96 pixels. Furthermore, the images will be transformed into grayscale, thus removing the RGB dimension. Additionally, the image will be processed with a Histogram of Oriented Gradients (HOG) transformer. Overall, for both the SVM and Random Forest Classifiers, the images will undergo grayscale transformation, HOG transformation, and rescaling before each pixel is entered into the model as a feature. This dataset was taken from Kaggle, entitled “Garbage Classification”. Images appear to be somewhat nonstandard, with different orientations, lighting, and distance from the object. This will be a challenge to train a classifier that can sift through the extra noise, however preprocessing steps will be taken to reduce noise. For the CNN, the preprocessing will occur on the training set, including random flips, zooms, and shears, to improve the model’s tolerance to noise.

4 Proposed Methodology

For a base model, a Support Vector Machine (SVM) will be applied. In multi-class classification, SVMs work by breaking down the problem into binary classification problems, using hyperplanes to distinguish the classes. Features (image pixels) will be rescaled, grayscaled, and undergo a histogram of oriented gradients (HOG) transformation to normalize the features to be input into the SVM. We believe this would be an effective base model, as it would produce a reasonable accuracy that can be improved upon and compared to other models. A grid search would then be applied to find the optimum hyperparameters of the data preprocessor, the HOG transformer. The orientation, cells per block, and pixels per cell would all be tuned to optimize performance.

Using our preprocessed images from the SVM, a Random Forest model will be applied. This model relies on decision trees, using entropy to maximize the information gain of the next branch or feature in traversing the tree to a class (leaf node). We believe that this model would improve upon our base model of the SVM, since it

is well suited to multiclass-classification. We also wanted to compare these two methods in their abilities to classify photos. Given the dataset’s uneven distribution of photos as well as the non-standard photos, lots of noise would be introduced and the Random Forest classifier may be better suited to handle the noise. The same grid search would be applied to optimize the hyperparameter choices in the HOG transformer preprocessor as well as hyperparameters specific to decision trees.

Finally, the SVM and Random Forest classifiers would be compared to the Convolutional Neural Network (CNN). We believe that this model will be the most robust model in classifying photos. Given the noise in the dataset, uneven distribution of photos, and the vast amount of data (photo dimensions), the CNN would be best suited to handle the massive amount of data and accurately identify the classes. Since tuning every single hyperparameter would be unfeasible given our computational power and time, the number of layers, convolutional filters, dropout rate, l1 and l2 regularization as well as learning rate will be tuned to maximize the effectiveness of our computational power and time. Early stopping will be applied as necessary.

5 Experimental Results and Analysis

5.1 SVM

After performing a HOG transformation on the grayscaled images, we tested classifier using stochastic gradient descent (SGD) with a maximum of 1000 epochs and a loss tolerance of $1e^{-3}$. This unoptimized model had an accuracy of 50%. To optimize the performance of this model, grid search with 3 splits of cross validation was used. For our parameter grid, we chose 2 sets of hogify orientations, 2 sets of hogify cells per block, and 3 sets of hogify pixels per cell. With this optimized model, it had an accuracy of 54%, implying that the accuracy increased slightly compared to the unoptimized model.

```
# Best parameters found:
{
    'hogify__cells_per_block': (2, 2),
    'hogify__orientations': 9,
    'hogify__pixels_per_cell': (8, 8)
}
```

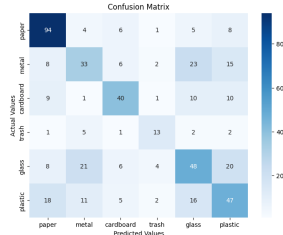


Figure 3: Confusion Matrix for SVM

	precision	recall	f1-score	support
paper	0.62	0.56	0.59	71
metal	0.46	0.45	0.45	107
cardboard	0.44	0.38	0.41	87
trash	0.68	0.80	0.73	118
glass	0.46	0.47	0.47	99
plastic	0.57	0.54	0.55	24
accuracy			0.54	506
macro avg	0.54	0.53	0.54	506
weighted avg	0.54	0.54	0.54	506

Figure 4: Classification Report for SVM

The prediction accuracy was the best for paper and suffered for all the other garbage labels. This is likely due to the following factors: large variation of paper images, little variation in each garbage label, an imbalanced dataset, and a small test size. For labels that have more variation in their images, this model would be more likely to classify them correctly, due to the property of the margin allowing for better generalization to diverse patterns and reducing the impact of noise in the data.

5.2 Random Forest

The optimized Random Forest model, achieved through the GridSearchCV has identified that a Random Forest classifier with 200 trees, no maximum depth, and requiring at least one sample to split a node yields the best cross-validated accuracy score (55.37%) on the training set. When applied to the test set, this model achieves a

slightly higher accuracy of 56.13%, indicating that it generalizes reasonably well to unseen data.

```
# Best parameters found:
{'classify__max_depth': None,
 'classify__min_samples_leaf': 1,
 'classify__n_estimators': 200}
#Training set accuracy:
55.37%
#Test set accuracy:
56.12%
```

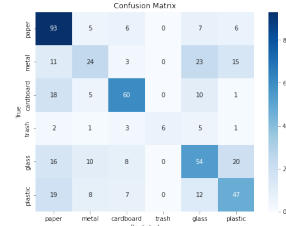


Figure 5: Confusion Matrix for Random Forest

	precision	recall	f1-score	support
paper	0.69	0.64	0.66	94
metal	0.49	0.50	0.49	108
cardboard	0.45	0.32	0.37	76
trash	0.58	0.79	0.67	117
glass	0.52	0.51	0.51	93
plastic	1.00	0.33	0.50	18
accuracy			0.58	506
macro avg	0.62	0.51	0.54	506
weighted avg	0.57	0.56	0.55	506

Figure 6: Classification Report for Random Forest

The confusion matrix (Fig 4) and classification report (Fig 5) give deeper insights into class-specific performance:

- The model shows high effectiveness in identifying 'paper' and 'trash', with true positive rates of 98.9% and 79.5% respectively. These materials also demonstrate the highest F1-scores.
- On the other hand, the precision for 'plastic' is 100%, signifying that every prediction made by the model for 'plastic' is correct, yet the recall is only 33%, denoting that it misses many actual instances of 'plastic'. This disparity suggests that while the

model is highly confident in its 'plastic' predictions, it is overly cautious, resulting in many false negatives.

- Conversely, 'cardboard' and 'glass' emerge as challenging materials for the model to classify accurately. 'Cardboard' has a notably lower recall and F1-score, with substantial misclassification with 'paper', indicating that the model struggles to differentiate between these two categories. Similarly, 'metal' and 'glass' are often confused with 'trash' and 'plastic'.

5.3 CNN

The CNN was able to produce a moderately accurate prediction, with a validation loss of 2.04 and validation accuracy of .7371. The prediction accuracy was best for cardboard, paper, and trash, and suffered for glass, plastic, and metal, likely due to a difficulty in differentiating such similar objects, and need for more features to train upon. While initial model parameters had caused underfitting, increasing layers and convolutional filters drastically induced overfitting. Batch normalization, dropout after dense layers, and l1 and l2 regularization were implemented to improve the generalization of the model. Furthermore, the model had a tendency to have an oscillating loss curve as it approached convergence. The learning rate had to be decreased, while the number of epochs and patience increased in order to allow the model to converge. Despite taking measures to prevent overfitting, the final model did show slight signs of overfitting, with the validation loss slightly above the training loss. The final training loss was 1.8717 and final validation loss was 2.0472. The model from epoch 50 was saved and implemented into the front end model. The learning rate was set to .001, l1 and l2 regularization was set to .001, and the dropout rate was .15 for the final model. It was set to train for 200 epochs with early stopping, patience being 20 epochs, the process stopped at epoch 70. Batch normalization was implemented after non-dense layers, and dropout was implemented after dense layers. Although higher accuracy was able to be attained, convergence of the model was

more important, so these parameters encouraged the model to converge and thus generalize well. Finally, the image preprocessing for the training data included horizontal flips, vertical flips, rescaling, shears, and zooms, with a .1 validation split (as was chosen to give the model ample training data). The preprocessor is shown below, and a batch size of 100 was chosen.

```
train = ImageDataGenerator(
    horizontal_flip=True, vertical_flip=True,
    validation_split=0.1, rescale=1./255,
    shear_range = 0.1, zoom_range = 0.1,
    width_shift_range = 0.1,
    height_shift_range = 0.1,)
test = ImageDataGenerator(rescale=1./
255, validation_split=0.1)
```

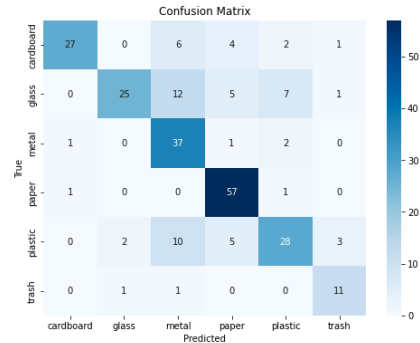


Figure 7: Confusion Matrix for Initial CNN

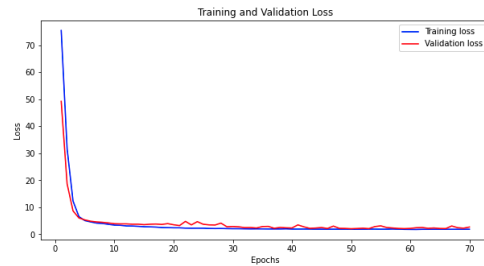


Figure 8: Loss for Initial CNN

6 Conclusion and Discussion

Through using a garbage classification dataset, which contains images of 6 different types of garbage and their labels, and three different models (SVM, Random Forest, CNN), we were able to build the following models: Support Vector Machine (SVM) with an accuracy of 54% Random Forest with an accuracy of 56% Convolutional Neural Network (CNN) with 3 convolutional 2D layers with an accuracy of 73%

Comparing the performance of the CNN with the other two classifiers, the CNN model demonstrated the highest accuracy on the testing dataset. We suspect that the CNN performed the best because of the ability of CNNs to learn hierarchical features and distinguish between intricate patterns with different scales and orientations. Since the label ‘trash’ had relatively small amounts of data compared to the other garbage labels, we could have improved the accuracy of the model by either removing or oversampling the ‘trash’ data. However, doing so would limit the real-world applications of the model because some garbage do not belong to the other five types of garbage. By retaining this label, it gives the model more flexibility with the types of garbage it can consider.

For our front-end implementation, we added an interface for a user to input an image of trash for our best model, which was our CNN, to classify. The uploaded image is resized to 98 pixels by 128 pixels, flattened to a vector, and normalized to values between 0 and 1. From this, the model can take this output and generate a set of probabilities for which of the six classes the image belongs to. We take the highest probability and display its corresponding label to the user as the final prediction.

References

Sources

- [1] National Overview: Facts and figures on materials, wastes and recycling — US EPA. (2023, November 22). US EPA. [https://www.epa.gov/facts-and-figures-about-materials-waste-and-](https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/national-overview-facts-and-figures-materials)

[recycling/national-overview-facts-and-figures-materials](https://www.epa.gov/facts-and-figures-about-materials-waste-and-recycling/national-overview-facts-and-figures-materials)

- [2] Aral, R. A., Keskin, Ş. R., Kaya, M., & Hacıömeroğlu, M. (2018). Classification of Trash-Net Dataset Based on Deep Learning Models. IEEE. <https://doi.org/10.1109/bigdata.2018.8622212>
- [3] Özkaya, U., & Seyfi, L. (2019). Fine-Tuning models comparisons on garbage classification for recyclability. arXiv (Cornell University). <https://arxiv.org/pdf/1908.04393>

Dataset

<https://www.kaggle.com/datasets/asdasdasdasdas/garbage-classification/data>